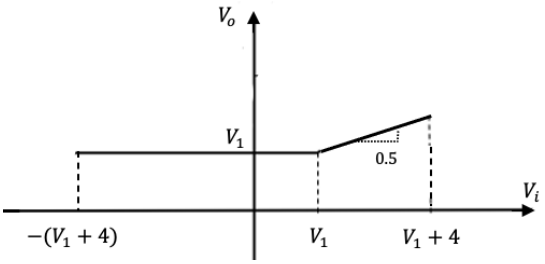
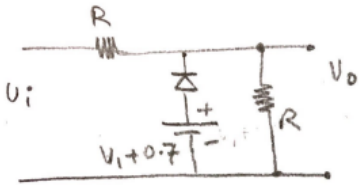
1) Design a diode limiter circuit to realize the transfer function shown below. Assume that the diodes have a 0.7-V drop when they conduct and that the input is a sinusoidal signal with amplitude as given in the figure. Specify the value of each component in your design. Explain your design.



One possible design is as below



R can have any value

when $U_i \leq V_1 \Rightarrow$ diode ON

and $U_o = U_i + 0.7 - 0.7$

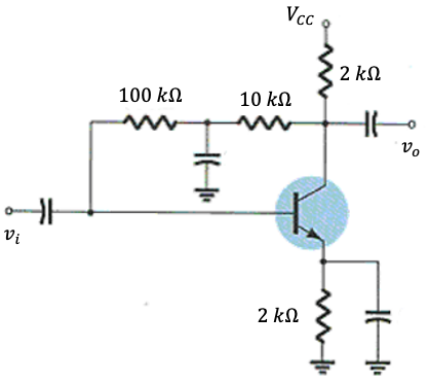
$= V_1$

when $U_i > V_1 \Rightarrow$ diode OFF

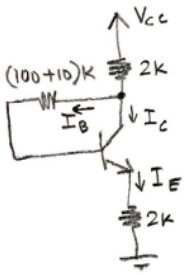
$U_o = \frac{R}{R+R} U_i = \frac{1}{2} U_i$

2) Analyze the circuit below and enter the answers in the table. Take β=80 and V_A=100 V. Also confirm that the transistor is in active mode. Show all steps and calculations.

Do you think this is a good voltage amplifier? How can it be improved? Explain in details.



DC Analysis



KVL

$V_{CC} - 2I_E - 110\,I_B - 0.7 - 2I_E = 0$

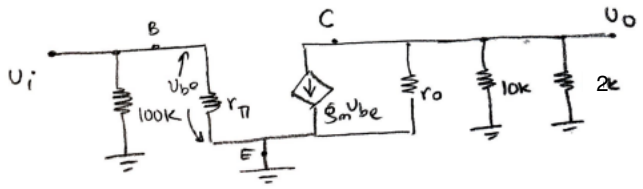
$V_{CC} - 0.7 = 434\,I_B \Rightarrow I_B = 434/(V_{CC}-0.7)$

$I_C = 80\,I_B \rightarrow V_C = V_{CC} - 2I_E$

$I_E = 81\,I_B \quad V_B = V_C - 110 I_B$

$V_{CB} = 110 I_B \Rightarrow$ active mode

AC Analysis



$v_i = v_{be}$

$v_o = -g_m v_{be} (r_o \| 10K \| 2K)$

$A_v = \frac{v_o}{v_i} = -g_m (r_o \| 10K \| 2K)$

$R_i = 100K \| r_\pi$

$R_o = r_o \| 10K \| 2K$

The input resistance is low, which is a negative aspect for a voltage amplifier. Remove the emitter bypass capacitor.

3) Implement the given logical function using CMOS logic. Perform transistor sizing such that the performance of the circuit matches the basic CMOS inverter with transistor size n and p. Find the total area of the circuit. Assume that the inputs and their complements are available. Show all details.

$F_1(A, B, C, D) = \overline{ABD} + (\overline{A + C})$

$F_2(X, Y, W, Z) = (\overline{X + W + Z}) \,.\, (\overline{Y + XZ})$

$F_3(K, L, M, N) = (\overline{KL + M}) \,.\, (KN)$

$F_1 = \overline{AB\,(D+C)}$

$F_2 = \overline{X + Y + W + Z}$

$F_3 = \overline{KL + M + \bar{K} + \bar{N}}$